

Grocery Delivery Management System

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INFO-C450 – Systems Design

Overview

The Grocery Delivery Management System is a Java-based application designed to help local grocery stores provide online ordering and delivery services. Many small grocery stores do not have affordable systems that allow customers to browse products, place delivery orders, and track deliveries.

This system solves that problem by offering a simple and efficient platform where customers can search for products, submit grocery orders, and monitor delivery status. Managers can maintain product inventory and assign delivery staff, while delivery staff can update order status during the delivery process.

The system improves efficiency, reduces manual errors, and helps grocery stores expand their services to customers who prefer online ordering.

This final version of the system includes expanded functionality, improved workflow design, and enhanced system documentation developed after the midterm phase.

Demo Link

https://1751071.kaf.kaltura.com/media/video1368678984/1_t17sdsq6

Project Repository

GitHub Repository:

<https://github.com/susaroge-svg/Grocery-Delivery-Management-System>

Key Features

User Registration and Login

Users can create accounts and securely log into the system.

Product Browsing and Search

Customers can view available grocery products and search by keyword or category.

Shopping Cart and Order Placement

Customers can add items to their cart and place grocery delivery orders.

Order Tracking

Customers can track the status of their orders as they move through the delivery process.

Product and Inventory Management

Managers can add, update, or remove grocery products and manage inventory levels.

Delivery Assignment

Managers can assign delivery staff to orders and monitor delivery progress.

Delivery Status Updates

Delivery staff update order status so customers receive accurate delivery updates.

System Requirements

To run the system, the following are required:

- Java Development Kit (JDK)
- NetBeans IDE
- Desktop computer running Windows, macOS, or Linux

The system uses **Java Swing for the user interface** and **in-memory data storage using Java ArrayLists**, so no external database installation is required.

System Architecture

The system consists of several main components:

- User interface screens for customers, managers, and delivery staff
- Application logic handling product management, order processing, and delivery updates
- An in-memory data storage component that stores products, users, orders, and deliveries

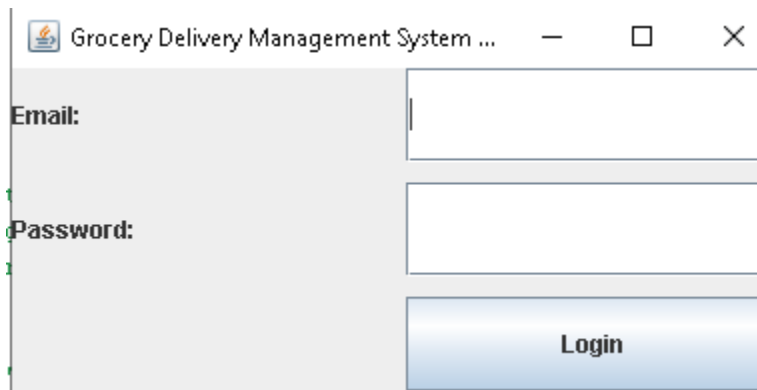
This architecture allows the system to remain lightweight and easy to deploy for small businesses.

Example Screens

(Insert screenshots from your NetBeans program if you have them)

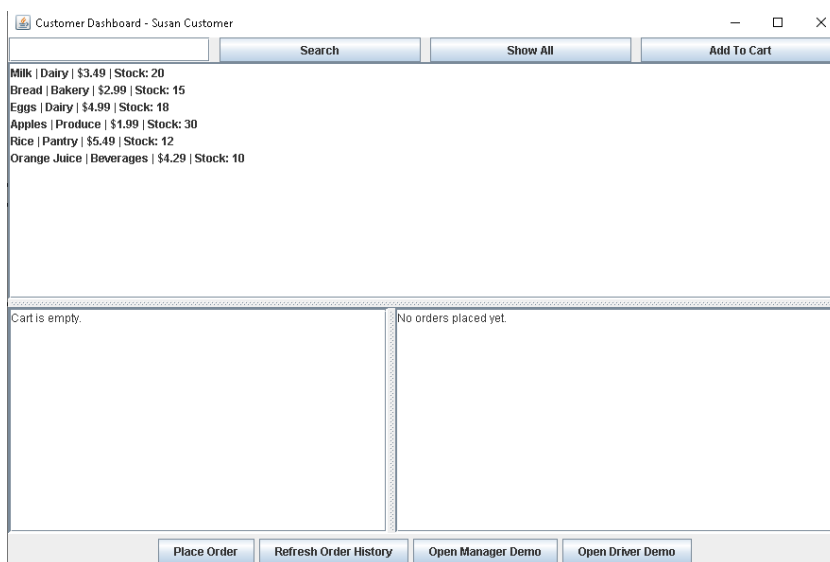
Example screens include:

- Login screen



The screenshot shows a window titled "Grocery Delivery Management System ...". It features a login form with two input fields: "Email:" and "Password:". Below the password field is a blue "Login" button. The form is set against a light gray background.

- Product browsing screen

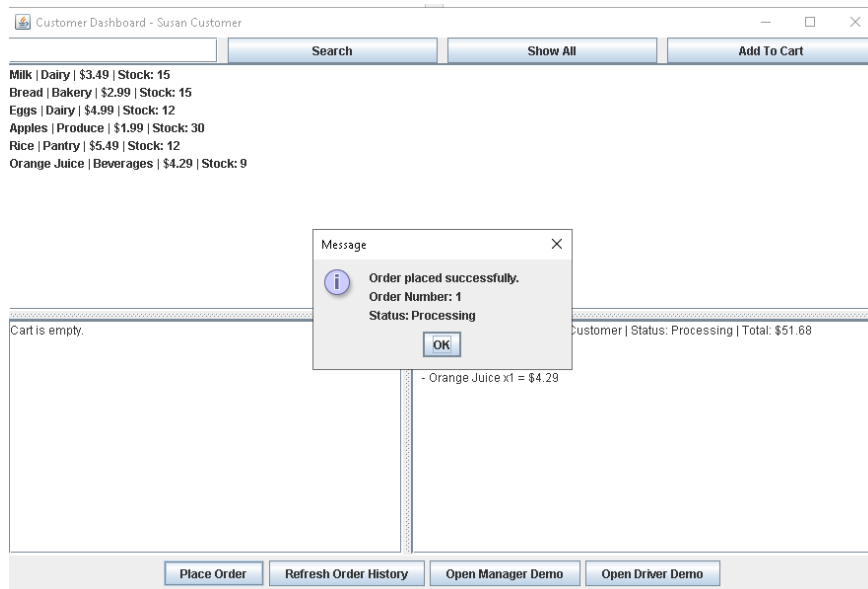


The screenshot shows a window titled "Customer Dashboard - Susan Customer". It features a product list with the following items:

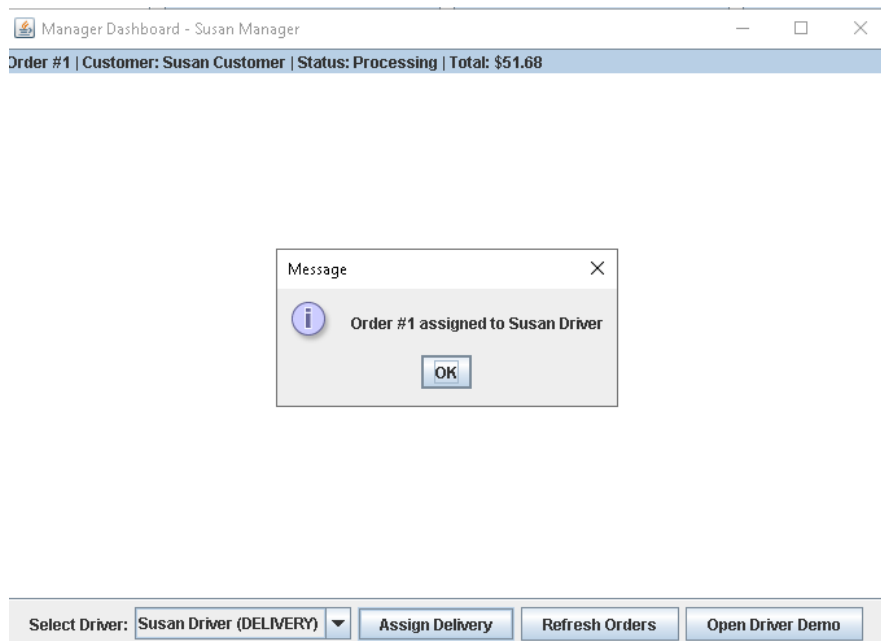
Product	Category	Price	Stock
Milk	Dairy	\$3.49	20
Bread	Bakery	\$2.99	15
Eggs	Dairy	\$4.99	18
Apples	Produce	\$1.99	30
Rice	Pantry	\$5.49	12
Orange Juice	Beverages	\$4.29	10

Below the product list, there are two sections: "Cart is empty." and "No orders placed yet." At the bottom of the window, there are four buttons: "Place Order", "Refresh Order History", "Open Manager Demo", and "Open Driver Demo".

- Shopping cart and order confirmation screen



- Manager dashboard for assigning deliveries



- Delivery status tracking screen

Delivery Dashboard - Susan Driver

Delivery #1 | Order #1 | Driver: Susan Driver | Status: Assigned

Message

Order #1 updated to: Delivered

OK

Mark Out for Delivery

Mark Delivered

Refresh